

NIDHIN D S

GENERATIVE AI / MACHINE LEARNING ENGINEER

RAG | AI AGENTS | LLM FINE-TUNING | NLP | PYTHON | FASTAPI | AWS

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SUMMARY

Generative AI and Machine Learning Engineer with hands-on experience in building end-to-end ML pipelines, NLP systems, and scalable AI applications. Skilled in Python, Scikit-learn, TensorFlow, and FastAPI, with growing expertise in Generative AI, Retrieval-Augmented Generation (RAG), LLM applications, and AI agents.

Strong understanding of the complete ML lifecycle, including data preprocessing, feature engineering, model development, evaluation, and deployment. Experienced in developing REST APIs for real-time inference and deploying applications on cloud platforms. Passionate about building scalable, production-ready AI solutions.

PROFESSIONAL EXPERIENCE

Data Science Trainer / Machine Learning Engineer

Techolas Technologies Pvt Ltd | Nov 2025 – Mar 2026

- Designed and implemented end-to-end machine learning pipelines for classification, regression, and NLP use cases
- Built and optimized models using Scikit-learn and TensorFlow, improving prediction accuracy and performance
- Developed NLP applications including sentiment analysis (LSTM) and text classification models
- Created FastAPI-based REST APIs for real-time model inference
- Led technical discussions on ML system design, evaluation metrics, and deployment strategies
- Collaborated with cross-functional teams to accelerate model experimentation and validation cycles

Technical Mentor / Machine Learning Engineer

LeSuccess Solutions (OPC) Pvt Ltd – Coimbatore | Jun 2025 – Oct 2025

- Designed and developed end-to-end ML workflows, including preprocessing, feature engineering, training, and evaluation
- Built predictive models using Scikit-learn and Python frameworks
- Developed NLP pipelines using NLTK and spaCy for text analytics
- Implemented production-style modular ML codebases simulating real-world deployment scenarios
- Led discussions on model architecture, performance metrics, and system design
- Guided hyperparameter tuning, validation, and optimization workflows
- Developed and deployed FastAPI-based ML APIs for real-time inference

Lecturer – Automobile Engineering

MGM Polytechnic – Kannur | Nov 2024 – May 2025

- Delivered instruction in automotive systems and vehicle dynamics
- Guided student projects involving data analysis and modeling
- Designed curriculum and assignments focused on real-world engineering applications
- Applied analytical and quantitative methods to explain complex concepts

Guest Lecturer – Mechanical Engineering

MGPTC Mananthavadi | Aug 2024 – Nov 2024

- Delivered lectures on core mechanical engineering subjects
- Simplified complex topics using structured problem-solving approaches
- Designed short-term modules aligned with industry-relevant applications
- Promoted data-driven and analytical thinking in engineering problems

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning & AI:

Scikit-learn, TensorFlow, Supervised & Unsupervised Learning, Model Training, Model Evaluation, Feature Engineering, Hyperparameter Tuning

Generative AI:

RAG (Retrieval-Augmented Generation), LLM Applications, Prompt Engineering, Hugging Face Transformers, AI Agents (Basic)

NLP:

NLTK, spaCy, Text Preprocessing, Tokenization, Lemmatization, Sentiment Analysis, Text Classification

ML Engineering & Deployment:

End-to-End ML Pipelines, FastAPI, REST APIs, Model Deployment, Real-Time Inference, Experimentation & Validation

Cloud & Tools:

AWS (EC2, S3), Docker (Basic), Streamlit, Render

Software Engineering:

Data Structures & Algorithms, OOP, Debugging, Code Reviews, Modular Code, SDLC

Collaboration & Tools:

Git, GitHub, Agile Methodology, Technical Documentation

PROJECTS

Telecom Customer Churn Prediction (End-to-End ML Project)

- Built a complete ML pipeline using Random Forest, achieving high classification performance
- Developed FastAPI APIs and integrated with Streamlit UI for real-time predictions
- Deployed on Render with version control using GitHub

California Housing Price Prediction (Regression Project)

- Developed regression model using XGBoost Regressor, achieving ~84.7% R^2 score
- Performed model comparison and hyperparameter tuning
- Built FastAPI APIs and deployed with Streamlit interface

BBC News Text Classification (NLP Project)

- Built NLP model using TF-IDF + Linear SVM for multi-class classification
- Implemented Scikit-learn pipeline with n-gram feature engineering
- Developed real-time inference APIs using FastAPI and deployed via Render

Enterprise RAG Assistant (Generative AI Project)

- Built RAG-based document Q&A system using FastAPI, FAISS, and Hugging Face
- Implemented semantic search + LLM-based response generation
- Developed REST APIs and deployed with Streamlit UI for real-time interaction

CORE COMPETENCIES

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|---|-----------------------------------|
| • Problem Solving & Analytical Thinking | • Model Optimization & Deployment |
| • Machine Learning System Design | • Handling Ambiguous Problems |
| • AI Application Development | • Cross-functional Collaboration |

EDUCATION

B.Tech – Mechanical Engineering

Cochin University of Science and Technology (CUSAT) | 2010 – 2014

B.Tech – Mechanical Engineering

Cochin University of Science and Technology (CUSAT) | 2010 – 2014

CERTIFICATIONS & TRAINING

Data Science / Machine Learning Training (Techolas Technologies)

Python, SQL, and Data Analytics Training Programs

ACHIEVEMENTS

- Successfully trained multiple batches in Data Science
- Improved student learning outcomes through real-world project training
- Developed structured curriculum for ML & Analytics programs

ADDITIONAL INFORMATION

- Languages: English, Malayalam, Hindi, Tamil
- Date of Birth: 01/04/1992